FurniCraft – Project Planning & Management

An Interactive 3D Furniture E-Commerce Platform with AI Assistance · Graduation Project · DEPI · October 2026

1. Project Proposal

Idea. FurniCraft is a furniture e-commerce platform that replaces static product images with interactive 3D visualization and customization. Users rotate and zoom furniture models, change colors, materials, and sizes, and see the preview and total price update instantly. An AI assistant built with prompt engineering helps users find products using natural language.

Problem. Online furniture shoppers struggle to visualize products accurately, compare customization options, and understand how materials and dimensions affect the final price.

Objectives.

- Build a modern, responsive furniture e-commerce experience.
- Provide real-time 3D product visualization and customization.
- Update previews and prices dynamically based on user choices.
- Integrate an AI assistant to simplify product discovery by recommending products.

Scope.

- **In scope (MVP):** 4 products at the start; catalog, categories, and advanced search with filters; product page with a 3D viewer (full rotation, zoom, color, material, and size changes, and live price); cart, authentication, and checkout (payment gateway in test mode or a mock); a simplified admin dashboard (products, customization options, and prices; orders; inventory; and simple analytics); and an AI assistant limited to recommending products.
- **Future enhancements, time permitting:** wishlist, product comparison, natural-language customization, AI-generated product descriptions, virtual room design, and Augmented Reality (AR).

Technology stack.

Layer	Technologies
Frontend	React.js, TypeScript, Bootstrap, React Router, Redux Toolkit, Three.js, React Three Fiber
Backend	Node.js, Express.js, RESTful APIs, JWT authentication
Database and storage	MongoDB, file storage for product assets and 3D models
AI	LLM API integration with prompt engineering

Expected outcome. A modern furniture e-commerce platform with interactive 3D visualization, configurable products, dynamic pricing, an AI-powered shopping assistant that recommends products, and a scalable foundation for future growth.

2. Project Plan

2.1 Milestones and deliverables

The dates are the DEPI deadlines.

Date	Milestone	Deliverables
16 Oct 2026	Planning and requirements	Project Planning & Management, Literature Review, Requirements Gathering
6 Nov 2026	System Analysis & Design	Problem statement, use case diagram, software architecture, ERD and schema, DFD, sequence, activity, state, and class diagrams, wireframes and UI/UX guidelines, technology stack, deployment and component diagrams
30 Nov 2026	Implementation	Source code on GitHub, branching strategy, README, deployed version or executable files
4 Dec 2026	Final presentation, testing, and reports	Test plan and test cases, bug reports, user manual, technical documentation, project presentation, infographic, optional demo video

2.2 Timeline (Gantt chart)

The development phases, as defined in the implementation plan of the proposal and updated after the lecturer's feedback:

1. Build a proof of concept with one 3D model (separate parts) and test changing its color and material; then build a complete 3D customization experience for one product (colors, materials, sizes, dynamic pricing).
2. Expand to the 4 products and connect to the database, catalog and filters, cart, authentication, checkout (test mode or mock), order management, inventory, and a simple admin dashboard.
3. Integrate the AI assistant for product recommendations.
4. Test, optimize, and document.

Phase	Start	End
Phase 1: 3D proof of concept and customization for one product	18 Oct 2026	6 Nov 2026
Phase 2: 4 products, database, cart, checkout, and order management	7 Nov 2026	22 Nov 2026
Phase 3: AI assistant for product recommendations	9 Nov 2026	27 Nov 2026
Phase 4: Testing, optimization, and documentation	23 Nov 2026	4 Dec 2026

2.3 Resource allocation

- **Team:** five members, with roles in Section 3.
- **Technology:** the stack listed in Section 1.
- **Code and documents:** the GitHub repository.

3. Task Assignment & Roles

Team member	Role
Rodina Mohamed	Team Leader; documentation and testing
Mohamed Gamal	3D
Ahmed Mohamed	Frontend
Amr Amer	Backend
Mohamed Ahmed	AI

4. Risk Assessment & Mitigation Plan

ID	Risk	Mitigation
R1	The AI assistant recommends products or options that do not exist	Validate every AI response against the catalog and allowed options before showing it; use few-shot examples and guardrails; show a clear fallback message
R2	Not enough time before the DEPI deadlines	Work phase by phase and finish the MVP first; move wishlist, comparison, natural-language customization, AR, and virtual room design to future work; review progress in a weekly team meeting
R3	3D models are hard to find or heavy, and their parts are not separate, so color and material cannot be changed per part	Decide the source of the models now; start with a proof of concept on one model; use light, optimized models with separate parts; test on a regular laptop and a phone

5. KPIs (Key Performance Indicators)

Metrics for project success, for example response time, system uptime, and user adoption rate.

KPI	How it is measured	Target
Page load time	Time until the main pages are usable	Under 3 seconds
3D viewer smoothness	Frame rate while rotating and zooming	At least 30 FPS on a regular laptop
Price update time	Time between changing an option and the new price appearing	Under 1 second
AI response time	Time from sending a request to receiving the answer	Under 5 seconds
AI output accuracy	Share of test prompts that return correct products from the catalog	At least 90%
Test pass rate	Test cases passed out of all test cases	At least 90%